

Graph Coloring.

Show that every planar graph has at ^{least one} vertex of degree ~~at most~~ 5 or less.

Proof:- (By contradiction)

Assume

G is planar and that all vertices of G have degree greater than or equal to 6.

$$\therefore 6v \leq 2e$$

$$\Rightarrow e \geq 3v$$

— (1)

According to

Euler's formula.

$$\cancel{v} \cdot e \leq 3v - 6 \quad \text{--- (2)}$$

$$\Rightarrow 3v \leq e \leq 3v - 6$$

impossible since $3v > 3v - 6$.

contradiction.

Edge Colourings:-

- A K -edge colouring α :

$$E_G \rightarrow [1, k] \text{ of a.}$$

graph G is an assignment of K colours to its edges.

G^α : G has the edge colouring α .

A vertex $v \in G$ and a colour $i \in [1, k]$ are incident with each other if

$$\alpha(vu) = i \text{ for some } u \in G.$$

if $v \in G$ is not incident with a color i , then i is available for v .

- The coloring α is proper, if no two adjacent edges obtain the same colour:

$$\alpha(e_1) \neq \alpha(e_2) \text{ for adjacent } e_1 \text{ and } e_2$$

The edge chromatic number

$\chi'(G)$ of G is defined as

$$\chi'(G) = \min \{ k \mid \exists \text{ a proper } k\text{-edge colouring of } G \}.$$

Five Color Theorem

Every planar graph is 5-colourable.

Proof: (Kemp's chain Argument)

The proof uses induction on the number of vertices v .

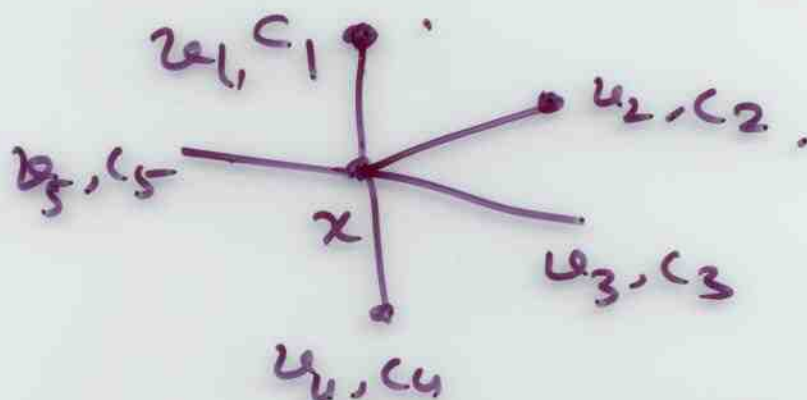
if $v \leq 5$ then G must be v -colourable, hence the Five-Color theorem holds.

Assume that the theorem holds for $v-1$.
Now there is a vertex x in G such that $d(x) \leq 5$.

If G' is formed by deleting vertex x from G together with the incident edges, then G' has $v-1$ vertices and can therefore, by the inductive hypothesis, be properly coloured with five colors. There are now two possible options: —

1) $d(x) < 5$ in which case x can be replaced and colored in one of the remaining colors which is not used by its adjacent vertices.

2) $d(x) = 5$. In this case either there is a spare colour for x (if two of its adjacent vertices in G are colored alike) or.



$c_i = \text{color of } u_i$

- i) there is no closed path connecting u_1 and u_3 so u_3 could be colored c_1 and x then be colored with c_3 .
- ii) there is a closed path connecting u_1 and u_3 , so all the vertices on the path can be coloured c_1 and c_3 alternatively. Since G is planar there can be no path connecting u_2 and u_4 such that all vertices on the path

are closed C_2 and C_4 , hence it can
be arranged for V_4 to be closed.
 C_2 and x can therefore be closed.
 C_4 .

Four color theorem

any map in a plane can be colored using four colors in such a way that regions sharing a common boundary do not share the same color.

- In 1852 F. Guthrie introduced this.
- Fallacious proofs were given by independently by Kempe (1879) and Tait (1880).
- In Dec 2004, G. Gonthier announced a proof.
- J. Freno (Peris. Comm. Nov. 8, 2005) gave short proof.